Applied Machine Learning Techniques For Predicting Economically Valuable Lumber Volume in Forests Using US Forest Service Data

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Response Variable Chosen: VOL.BF

Study Type Chosen: Regression Problem

VOL.BF (Net Board-Foot Volume): A "board foot" is a specialized unit representing a piece of lumber 12" x 12" x 1". This metric estimates how much usable lumber can actually be sawn out of the tree after accounting for the "kerf" (the sawdust lost to the saw blade) and the round shape of the log.

INTRODUCTION

The US Forest Service is an agency of the United States Department of Agriculture, administering 154 national forests. One of the most applicable ways to use machine learning in the corporate world is for helping to estimate economic outcomes or forecast revenues. By using data provided by the US Forest Service, machine learning methods can be applied to accurately predict response variable VOL.BF, the net board-feet volume estimate of how much usable lumber can be sawn from a tree and used for products such as for houses, furniture, etc. Therefore, predicting the volume of usable lumber can be especially important in corporate finance, providing lumber companies with accurate predictions to increase and forecast future lumber production, revenue, and the net present value of a forest tract (how much a plot of forest is actually worth in current time). Additionally, Timberland Investment Management Organizations (TIMOs) or Real Estate Investment Trusts (REITs), companies that are specialized asset managers that acquire, manage, and sell forestland portfolios for institutional investors can especially benefit from the goal of this study. Even if national forests are protected land, the data provides a strong foundation for creating predictive models to be used on other unprotected forests for harvesting wood or investment. Therefore, I believe a regression problem is highly suitable for the desired context and importance of predicting VOL.BF using US Forest Service data.

The data for this study comes from the Forest Inventory and Analysis (FIA) program within the US Forest Service. The FIA program is an organization of in-field foresters who measure and monitor key metrics related to the extent, condition, and trends of all forests in the United States, thus producing and sharing data used to analyze forest resources. The dataset used in this study provides six survey variables that can be used as response variables and they can be found listed in the appendix.

The response variable of interest that will be used in my study, as stated, is VOL.BF. The remaining response variables will be ignored and removed from the data for the purpose of this study. There are 25 total variables, 6 response variables, 18 predictor variables, and 8752 observations included in the full dataset. There is also a train indicator variable used to separate the data into training and testing subsets, where the indicator equals 1 for the training data split and equals 0 for the testing data split. The total broader population related to the data is the collection of all possible forest land plots across the 154 national forests in the US. The 18 explanatory/predictor variables included in the dataset can be found listed in the appendix. The numeric explanatory variables provide several measurements for the forest land representing land topography, climatic conditions, vegetation, and wildfire risk. Each observation included in the dataset corresponds to a 90x90 meter "pixel" of land at a random sample of point locations in the Western US. Therefore, the population for this data could be considered as all possible 90x90 meter pixels of land in Western US. Each observation or pixel of land is identified by their section ID (M333A, M333B, M333C, or M333D).

VOL.BF can be considered in a classification problem by classifying the variable into four classes indicating the sawtimber harvestability of trees for the use of and quality to be processed into sawn lumber, veneer, or planks. The bins and classes associated considered are as follows:

Bin Quantities	**Bin Name**	**Class Numeric Indicator**
VOL.BF = 0	Non-Sawtimber	0
0 < VOL.BF ≤ 150	Small Sawtimber	1
150 < VOL.BF ≤ 400	Standard Sawtimber	2
400 < VOL.BF	Large Sawtimber / Old Growth	3

However, this study will focus on VOL.BF in a regression problem, keeping VOL.BF as a numeric response variable without being binned. The reasoning for this decision is for the purpose of my own learning development in applying machine learning models to quantified predictions in a real-world scenario. Treating VOL.BF as a quantitative response in a regression problem is also suitable in regards to the importance of the study to accurately predict and quantify the economic value of a plot of forest land. Therefore, the goal of this study is to create three distinct machine learning models to predict VOL.BF in a regression situation.

Three machine learning methods that will be used to build accurate predictive models in order to estimate VOL.BF on new unmeasured pixels of land. The machine learning models will therefore be fit to the training data subset, then will be tested and scored based on performance for making predictions on VOL.BF in the testing data subset. The models will be scored based on their mean squared error (MSE) and, for interpretability, root mean squared error (RMSE). The three models that will be used for prediction are Lasso Regression, the Random Forest, and the Deep Neural Network.

EXPLORATORY DESCRIPTIVE ANALYSES

A correlation matrix, pairwise scatterplots, variable histograms, and VIF estimates were used for conducting exploratory descriptive analyses. According to the correlation matrix and pairwise scatterplots, all PRISM temperature variables are highly correlated with each other, given that they each report statistics on temperature such as mean, maximum, and minimum annual temperature. Therefore, including all the PRISM variables in a model would be redundant and introduces the issue of multicollinearity. Additionally, variables for measures of precipitation or vapor, such as vpdmax, ppt, and def are strongly correlated. Ndvi and evi are both LANDFIRE statistics and are highly correlated and collinear. VIF scores and variable selection methods will be used to eliminate redundant variables and solve for any collinearity.

The pairwise scatterplots are split into three groups for easier visualization and are also color coded by the variable SECTION. They can be seen in the three figures below:

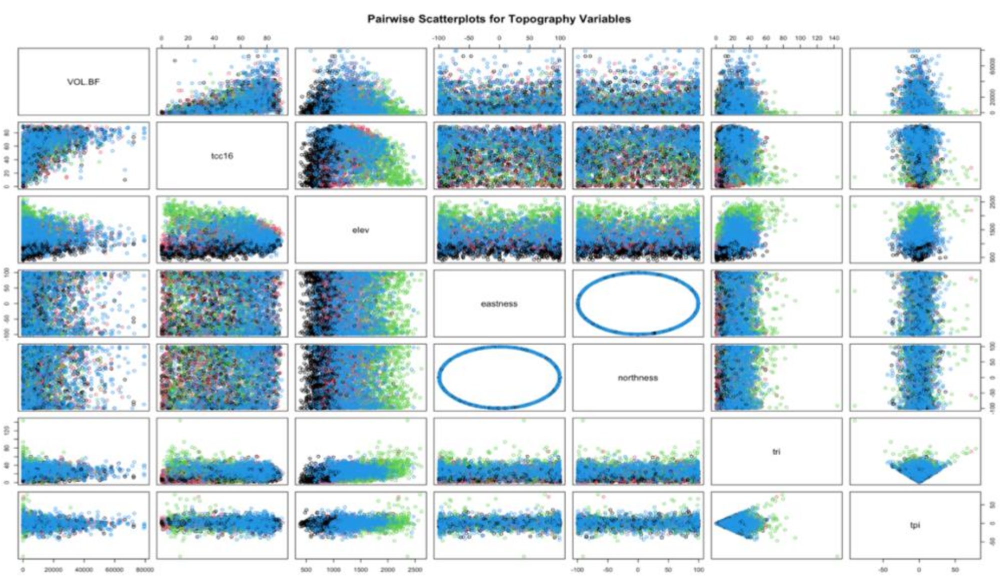

FIGURE 1 *Pairwise Scatterplots for Topography Variables*

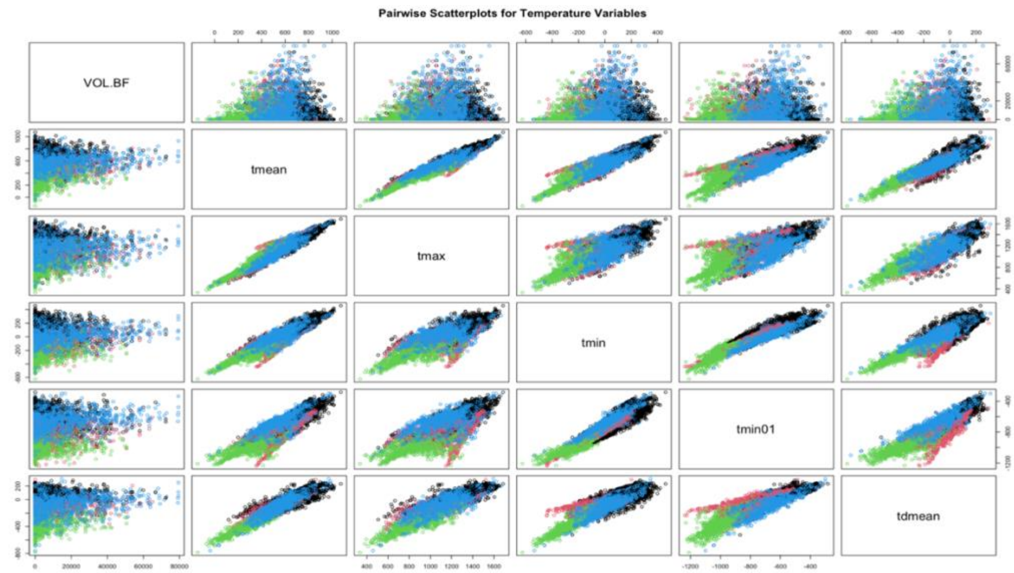

FIGURE 2 *Pairwise Scatterplots for Temperature Variables*

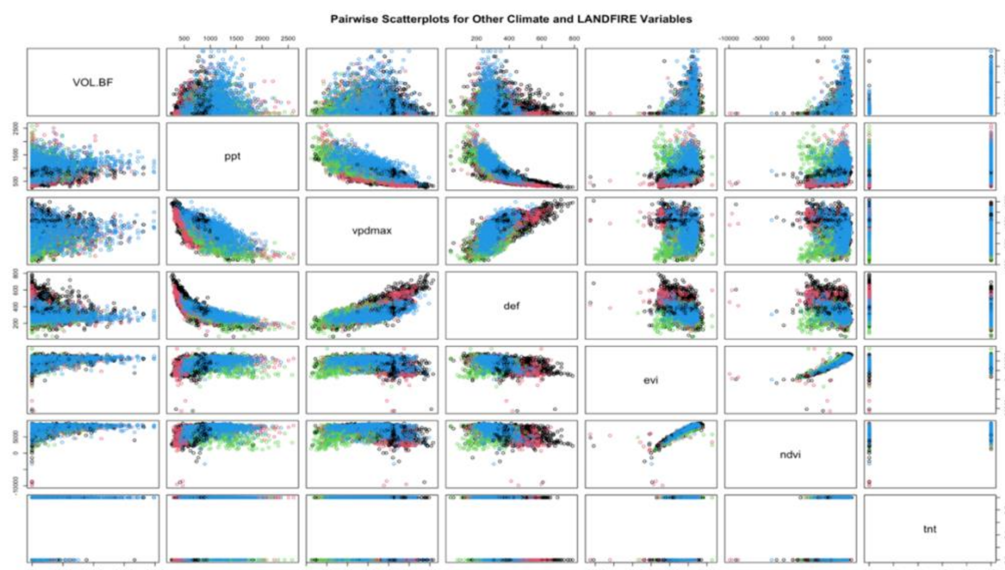

FIGURE 3 *Pairwise Scatterplots for LANDFIRE and Other Climate Related Variables*

When color coding for SECTION in the pairwise scatterpots, there does not appear to be any strong cases evident for an interaction between two variables, therefore there will be no use of interaction terms. There also does not appear to be any significant cases of non-linear relationships between any predictor and VOL.BF, therefore no higher degree terms will be used.

According to the category counts for SECTION and histograms of variables, there does not appear to be any concerns for sparsity in the data. The histograms can be seen in Figure 4 below:

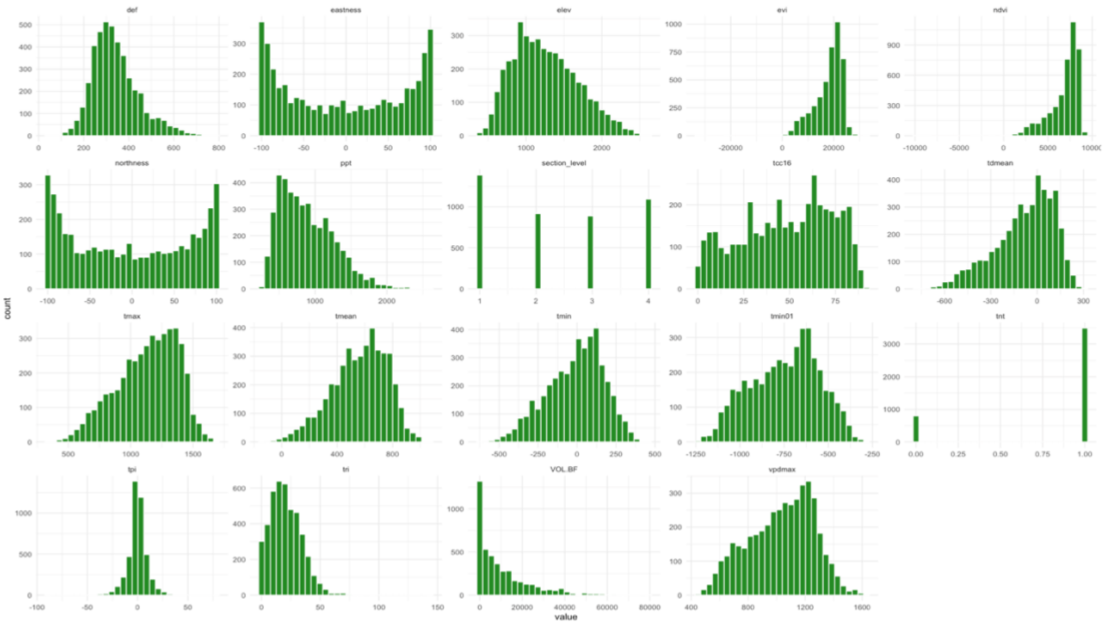

FIGURE 4 *Histograms for All Variables. The histograms are used to check for sparsity in the data.*

A simple baseline intercept-only baseline regression model is fitted and included to provide a baseline minimum MSE and RMSE to improve upon. The results of the baseline model can be found in Table 1:

	Baseline
(Intercept)	10364.024***
	(194.744)
Num.Obs.	4286
R2	0.000
R2 Adj.	0.000
AIC	93199.3
BIC	93212.0
RMSE	12747.90

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

TABLE 1 *The Intercept and RMSE for the Baseline Intercept-Only Model for Predicting VOL.BF*

The baseline intercept-only model reported an intercept of 10364, meaning the average for VOL.BF using the training data is 10364 board-feet in sawlog portion of a sawtimber tree.

MODEL PROCEDURES AND RESULTS

Before fitting the three models mentioned, a baseline model, full model, and three models using backward, forward, and best subset selection methods are fitted and the results are shown in Table 2:

	Intercept Only	Full	Backward Selection	Forward Selection	Best Subset Selection
(Intercept)	10364.024***	-18585.816**	-8598.921***	-9714.365***	-13325.428**
	(194.744)	(7064.645)	(2108.453)	(1503.435)	(4078.129)
SECTIONM333B		849.801+	986.801*	937.785*	940.967+
		(508.014)	(478.878)	(474.472)	(480.533)
SECTIONM333C		-414.531	-104.748	-233.447	-156.369
		(696.415)	(624.780)	(601.135)	(642.900)
SECTIONM333D		1551.643**	1560.764**	1497.809**	1811.157***
		(559.802)	(479.892)	(472.577)	(501.224)
tcc16		312.511***	314.916***	314.616***	314.820***
		(11.547)	(10.919)	(10.911)	(10.910)
elev		1.637			
		(1.689)			
eastness		4.959*	5.085*	5.109*	5.129*
		(2.221)	(2.214)	(2.214)	(2.214)
northness		-6.078*	-5.681*	-5.753*	-5.637*
		(2.395)	(2.332)	(2.330)	(2.334)
tri		45.568**	43.551**	45.409**	44.972**
		(14.369)	(14.190)	(13.974)	(14.067)
tpi		-63.870***	-58.836***	-58.301***	-58.813***
		(17.035)	(16.272)	(16.256)	(16.283)
ppt		4.410***	4.365***	4.542***	4.675***
		(0.918)	(0.693)	(0.653)	(0.797)
tmean		-119.389		6.304***	14.600**
		(391.295)		(1.410)	(4.922)
tmax		73.670	2.304+		
		(195.589)	(1.326)		
tmin		67.199	4.222**		
		(195.744)	(1.582)		
tmin01		-4.856			-4.556
		(3.412)			(3.212)
vpdmax		-7.415			-4.224+
		(6.300)			(2.295)
tdmean		-1.089			
		(4.567)			
def		-2.514			
		(3.502)			
tnt		-1584.388**	-1555.038**	-1558.592**	-1507.678**
		(560.919)	(556.193)	(556.144)	(557.100)
evi		0.030			
		(0.075)			
ndvi		-0.547*	-0.424**	-0.421**	-0.470**
		(0.236)	(0.158)	(0.158)	(0.160)
Num.Obs.	4286	4286	4286	4286	4286
R2	0.000	0.351	0.350	0.350	0.350
R2 Adj.	0.000	0.348	0.348	0.348	0.348
AIC	93199.3	91388.5	91379.6	91378.2	91378.7
BIC	93212.0	91528.5	91475.1	91467.3	91480.5
RMSE	12747.90	10272.32	10278.49	10279.18	10274.96

TABLE 2 *Coefficients and Output Error Metrics for Baseline, Full, BS, FS, and BSS Models*

Lasso is also fitted using all 18 predictor variables, before removing any collinear variables, and the coefficients are shown in the Table 3:

	s=24.77076
(Intercept)	-6652.9927299
SECTIONM333B	821.8235266
SECTIONM333C	-68.4418557
SECTIONM333D	1592.7989351
tcc16	308.5080094
elev	0.0000000
eastness	4.5966425
northness	-5.5499920
tri	41.7021477
tpi	-56.9874231
ppt	3.6989326
tmean	3.6287294
tmax	0.0000000
tmin	2.2457071
tmin01	0.0000000
vpdmax	0.0000000
tdmean	0.6774602
def	-2.2713066
tnt	-1486.5976815
evi	0.0000000
ndvi	-0.3747790

TABLE 3 *The Coefficients for Lasso Regression Using All 18 Explanatory Variables*

Notice that Lasso shrinks coefficients to zero for collinear variables, elev, tmax, tmin01, evi, and vpdmax, leaving 13 variables left. After conducting EDA and reviewing selection methods from Lasso, BS, FS, and BSS, 6 redundant collinear variables were selected to be removed from the data: tmin, tmax, tmin01, tdmean, ndvi, and vpdmax. After removing the variables, Lasso now reports coefficients as follows using the 12 remaining predictor variables:

	s=0.6579332
(Intercept)	-1.118109e+04
SECTIONM333B	8.932392e+02
SECTIONM333C	-2.151436e+02
SECTIONM333D	1.329234e+03
tcc16	3.071417e+02
elev	9.557834e-01
eastness	4.842253e+00
northness	-6.029404e+00
tri	4.680359e+01
tpi	-6.156710e+01
ppt	3.863075e+00
tmean	8.569801e+00
def	-3.482551e+00
tnt	-1.822133e+03
evi	-8.485558e-02

TABLE 4 *The Coefficients for Lasso Regression Using Subset of 12 Explanatory Variables*

After removing the 6 collinear variables, the three models, Lasso, Random Forest, and Deep Neural Network were fitted and used for prediction on the testing dataset (Ridge and an RF-DNN average were also fitted, but are not included as any of the three models focused on for this study). Their respective MSE and RMSE scores are reported in Table 5:

Model Performance Comparison		
Model	**Test MSE**	**Test RMSE**
Ridge	106608358	10325.13
Lasso	106406248	10315.34
Random Forest	103585639	10177.70
DNN	102390926	10118.84
RF-DNN Avg.	100586609	10029.29

TABLE 5 *Performance Metrics for Ridge, Lasso, Random Forest, DNN, and RF-DNN Prediction Avg.*

The baseline model scored a test MSE and RMSE of 163898034 and 12802 respectively. Because the scaling of VOL.BF is so large, RMSE is used as a more interpretable metric. The baseline model reports an RMSE of 12802, meaning on average, the prediction for VOL.BF has an error of around 12802 board-feet. The range of VOL.BF in the test data is 85885, therefore the baseline model prediction error is about 14.9% of the total range of VOL.BF. The full model with all predictors included reports a test MSE of 106618068 and RMSE of 10325.

As can be seen in the table above, Ridge and Lasso Regression report similar results while Random Forest performs better than Ridge and Lasso and the Deep Neural Network performs best overall with the lowest MSE, and RMSE reported as 10118, about 11.78% of the full range of VOL.BF. Overall, the DNN reduced the error from 12802 (Baseline) to 10118 (DNN), indicating a 20.9% reduction in error or a reduction of 2684 board-feet. Therefore, the DNN model performs significantly better than the baseline model.

Cross validation was utilized during Ridge and Lasso Regression to select the optimal value for the complexity penalization parameter lambda. Using 12 predictors, Ridge and Lasso regression each reported test RMSE of 10325 and used optimal lambdas of 43.29 and 0.66 respectfully. Lasso uses all predictors remaining after removing redundant predictors. However, if redundant predictors are not removed, Lasso automatically shrinks cofficients of insignificant predictors to zero and uses 13 predictors, some of which are different from my selected 12. Lasso reports a slightly lower RMSE of 10315 if it has access to all 18 predictors to choose from.

500 trees were used in constructing the Random Forest model. Additionally, m = 4 predictors were selected when randomly sampling a subset of predictors to use for each tree (12 predictors → 12/3 = 4 in

subset). After tuning Random Forest using OOB Error, the optimal m = 3 predictors were used to fit Random Forest and a lower RMSE of 10177 is obtained.

The Deep Neural Network model performed the best in the end with the lowest RMSE of approximately 10119. DNN used 50 epochs and 64 input neurons with two hidden layers. The first hidden layer included 32 neurons and the second hidden layer included 16 neurons. The RMSE 10119 corresponds to a model prediction error that is about 11.78% of the full range of VOL.BF.

Overall, the simple baseline model performed with a significantly higher error than the three models. Deep Neural Network is the strongest performing model and provides the best results with lowest error, but it requires a high amount of computing power and time to run. An average of the predictions between RF and DNN provides an even lower RMSE estimate and higher accuracy for prediction! The RMSE reported using the RF-DNN average is 10029, just 11.68% of the range of VOL.BF.

CONCLUSION

The regression framework successfully predicted wood volume VOL.BF, improving substantially upon the simple baseline model. The baseline model reported a high initial RMSE of 12802 board-feet, representing 14.9% of the total response range. All three advanced machine learning techniques proved to be better predictive tools than the baseline model. Lasso Regression achieved a RMSE of 10325 using the subset of 12 selected predictors. The Random Forest improved further upon Lasso by capturing non-linear interactions, dropping the RMSE to 10177. The Deep Neural Network performed the best of all models, achieving the lowest RMSE of 10119, representing a 20.9% reduction in error relative to the baseline model. The Deep Neural Network prediction error is 11.78% of the total range, indicating a successful regression and adequately dependable model.

Each model procedure also had distinct limitations. Lasso regression assumes linearity, therefore limiting it from including complex, high-order interactions and capturing any non-linear trends. Variables were scaled prior to conducting Lasso/Ridge regression. The benefits to fitting the Random Forest is the ability to make precise predictions while reducing variance and balancing variance/bias through the use of ensemble learning, using numerous trees and selection of random sample of predictors for each. The Random Forest is also not constrained to linearity, and can fit a complex model. However, Random functions in a way which does not provide any equation or visualization for interpretability. Additionally, Random Forest is based on trees, therefore it cannot extrapolate beyond the minimum and maximum boundaries of the training data. The DNN achieved the highest accuracy, but is computationally expensive, requires significant processing time, and is highly sensitive to parameter configurations. DNN also offers no interpretability about the model and how its inputs translate to the final output value.

For the US Forest Service and companies who profit from forests, the results of this study provide marvelous strategic value for investing in and managing forest land. The removal of the 6 collinear variables provide a more concise set of 12 predictors to create accurate predictive models. Furthermore, forest structural dynamics are non-linear and variant such that relying on simple linear models do not provide enough complexity to be accurate about estimating VOL.BF in new unanalyzed forests. The DNN model provides a 20.9% reduction in test error, translating to more accurate valuations of forest land pixels and estimation of harvestable timber, increasing the revenue and lowering risk for firms such as lumber companies, TIMOs, and REITs.

APPENDIX

Survey Response Variables:

BA	Basal area of trees (DIA * DIA * 0.005454) (sqft)
VOL.CF	Net cubic-foot volume in sawlog portion of a sawtimber tree
VOL.BF	Net board-foot volume in sawlog portion of a sawtimber tree
COUNT	Number of trees per acre
DRYBIO	Biomass in aboveground portion of tree (lbs)
CARBON	Carbon in aboveground portion of tree (lbs)

Auxiliary Explanatory Variables:

Quantitative:

tcc16	National Land Cover Dataset, Analytical Tree Canopy Cover
elev	LANDFIRE 2010 DEM - elevation, in meters
eastness	Transformed aspect degrees to eastness (-100 to 100)
northness	Transformed aspect degrees to northness (-100 to 100)
tri	Terrain Ruggedness Index
tpi	Topographic Position Index
ppt	PRISM mean annual precipitation - 30yr normals (1991-2020) (mm*100)
tmean	PRISM mean annual temperature - 30yr normals (1991-2020) (degC*100)
tmax	PRISM mean annual maximum temperature - 30yr normals (1991- 2020) (degC*100)
tmin	PRISM mean annual minimum temperature - 30yr normals (1991- 2020) (degC*100)
tmin01	PRISM mean minimum temperature (Jan) - 30yr normals (1991-2020) (degC*100)
vpdmax	PRISM max annual vapor pressure deficit - 30yr normals (1991-2020) (Pa*100)
tdmean	PRISM mean annual dewpoint temperature - 30yr normals (1991-2020) (degC*100)
def	TOPOFIRE mean annual climatic water deficit - 30yr normals (1981-2010)
evi	LANDFIRE 2010 Landsat - (2.5*((b4-b3)/(b4+6*b3-7.5*b1+1))) * 10000)
ndvi	LANDFIRE 2010 Landsat - (((b4-b3)/(b4+b3))*10000)

Categorical:

SECTION	Geographical region, based on ecological and geological features.
tnt	LANDFIRE 2014, landsat-based (1/0: tree/non-tree)